Part 1: Electron Transport Chain

Which of the following is NOT a function of ATP?

A) Muscle contraction

B) Nerve conduction

C) Storage of genetic information

D) Active transport across membranes

ANSWER: C

Substrate-level phosphorylation is characterized by:

A) Dependence on cellular oxygen supply

B) Rapid energy production independent of oxygen

C) Occurrence only in mitochondria

D) Involvement of the respiratory chain

ANSWER: B

Which complex is also known as NADH:CoQ oxidoreductase?

A) Complex I

B) Complex II

C) Complex III

D) Complex IV

ANSWER: A

How many electrons are required to reduce one atom of oxygen?

A) 1

B) 2

C) 3

D) 4

ANSWER: B

$FADH_2$ enters the electron transport chain at:

A) Complex I

B) Complex II

C) Complex III

D) Complex IV

ANSWER: B

Which complex does NOT pump protons?

A) Complex I

B) Complex II

C) Complex III

D) Complex IV

ANSWER: B

The enzyme that catalyzes the transfer of electrons from cytochrome c to oxygen is:

A) NADH dehydrogenase

B) Succinate dehydrogenase

C) Cytochrome c oxidase

D) ATP synthase

ANSWER: C

How many ATP molecules are produced from the oxidation of one NADH?

A) 1.0

B) 1.5

C) 2.0

D) 2.5

ANSWER: D

How many protons must enter through ATP synthase to produce one ATP?

A) 1

B) 2

C) 3

D) 4

ANSWER: D

Which complex in the electron transport chain is inhibited by Rotenone?

A) Complex I

B) Complex II

C) Complex III

D) Complex IV

ANSWER: A

Which complex in the electron transport chain is inhibited by cyanide?

A) Complex I

B) Complex II

C) Complex III

D) Complex IV

ANSWER: D

Which of the following is an uncoupler of oxidative phosphorylation?

A) Rotenone

B) Antimycin A

C) Dicoumarol

D) Cyanide

ANSWER: C

Part 2: Free Radical Toxicity

Free radicals are highly reactive due to:

A) Full outer shell

B) Presence of unpaired electron(s)

C) Absence of electrons

D) Stable electronic configuration

ANSWER: B

Which of the following is NOT a source of free radicals?

A) Environmental toxins

B) Normal metabolism

C) Inert gases

D) Radiation

ANSWER: C

Ozone (O_3) is:

A) A free radical

B) A powerful oxidizing agent

C) An antioxidant

D) An inert gas

ANSWER: B

Superoxide dismutase (SOD) is an example of:

A) Primary antioxidant

B) Secondary antioxidant

C) Tertiary antioxidant

D) Pro-oxidant

ANSWER: A

Oxidative stress is defined as:

A) Balance between oxidants and antioxidants

B) Imbalance between oxidant free radicals and antioxidant systems

C) Complete absence of free radicals

D) Increase in antioxidant levels only

ANSWER: B

Ischemic reperfusion injury occurs due to:

A) Decreased oxygen supply only

B) Generation of oxygen free radicals upon restoration of blood flow

C) Increased antioxidant activity

D) Blockade of Complex I

ANSWER: B

Which of the following hematological diseases is associated with free radicals?

A) Hemophilia A

B) Favism

C) Von Willebrand disease

D) Thalassemia trait

ANSWER: B

Free radical damage accumulates with:

A) Exercise

B) Age

C) Antioxidant intake

D) Hydration

ANSWER: B